

Rendered, Not Transported: Epistemic Disposition in Multi-Agent LLM Pipelines

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Abstract

Multi-agent LLM pipelines route work through specialist models and aggregate their outputs through a final writer. When epistemic qualifications—hedges, labelled assumptions, cutoff caveats—fail to reach the user, the prevailing diagnosis is a *transport* failure: uncertainty is said to be erased at agent interfaces, motivating proposals to attach uncertainty-bearing carriers alongside the text. We present evidence that this diagnosis misplaces the fault. Holding the pipeline, the models, and the upstream specialist text fixed, and changing *only* the final writer’s instructions, moves output disposition by $3\text{--}4\times$ ($0.16 \rightarrow 0.53$ as preservation clauses are restored; $0.20 \rightarrow 0.64$ from a single clause). The qualifications were present in the transmitted text the whole time. What varied was whether the last writer *rendered* them. Disposition is a rendering property of the final writer, not a transport property of the pipeline.

We characterise the last writer’s **rendering function** across $\sim 1,300$ pre-registered audited runs of a four-model “Council of Experts” on consumer hardware, and report four properties. (1) *Structure is not the lever*. Holding the model fixed across six architectures (single-shot, flat merge, two-round debate, self-refine, prompted single model, council), multi-agent topology lifts disposition negligibly (flat merge 0.16 vs. single-shot 0.14; two-round debate 0.16), and a single prompted model matches the full council on magnitude (0.50 vs. 0.57, overlapping CIs). What the council uniquely achieves is *calibration*: it is the only architecture combining substantial disposition with a clean trigger-free control (0.00 [0.00, 0.00] vs. the prompted model’s 0.15 [0.08, 0.22], disjoint), because its instruction is *conditional* on evidence that actually exists upstream. (2) *Upstream weights are not the lever*. Prompting and exemplar SFT produce large seat-level lifts ($\approx 2\times$) that are indiscriminate and vanish at the pipeline’s mouth; reference-free preference optimization installs no seat-level magnitude across three lineages, invariant to $3.2\times$ dose. Its one real effect is subtractive and conditional—suppressing unwarranted hedging on a weakly-aligned seat—which survives triangulation across three instruments and generalizes across ORPO and CPO. Triangulation also caught and retired one finding produced by the primary metric alone. (3) *Downstream weights are not the lever either*. Training the final writer directly leaves its band unmoved, and so does an on-policy retry whose reward is calibration-shaped and computed at the pipeline’s mouth—the strongest feasible analogue of multiturn-aware reward training. Weight-level training fails at every locus tested, under every reward design tested. (4) *Instructions are the lever, and they are graded and targeted*. Output density rises with the number of preservation clauses on all three writers (Spearman $\rho = +1.00, +0.80, +0.80$; a monotone trend, strictly monotone only on gpt-oss). Isolating clauses shows each lifts *exactly* the behavior family it names and no other—these are targeted specifications, not general priming—while the composite gain is carried by whichever single clause matches the family that writer already produces in volume.

Finally, the writers’ *intrinsic* bands are indistinguishable: four models spanning four architectures all emit 0.14–0.15 under a neutral prompt. Models therefore do not differ in disposition; they differ in **instruction gain**—how strongly each amplifies a disposition instruction. This relocates the mechanism from a characteristic emission level to a per-model coefficient on instructions, and explains why weight training could not move it: a LoRA-scale preference run has no purchase on how a model processes instructions. Practically: select a final writer for instruction responsiveness, then write its instructions

conditionally over real upstream evidence. All experiments are \$0-budget and reproducible on a single 32 GB consumer machine; pre-registrations, protocol amendments, refuted hypotheses, and a post-hoc integrity audit that withdrew one arm are included in the record.

1 Introduction

Production LLM systems increasingly take the form of pipelines: multiple models, often specialized, whose outputs are aggregated by a final model into the answer a user sees. A recognised failure of such systems is that epistemic qualification does not survive the trip. A specialist writes “assuming 60% adoption” or “this may have changed since my training data”; the user reads a confident synthesis with the qualification gone.

Transport or rendering? The dominant framing treats this as a **transport** problem. Uncertainty is cargo; agent interfaces transmit discrete commitments rather than epistemic states; the cargo is therefore erased in transit. The remedies follow from the diagnosis: propagate numeric confidence through the chain [27], or attach an uncertainty-bearing latent carrier to each handoff because text cannot hold the freight [26]. On this view the fix is architectural, requires access to model internals, and is unavailable for black-box components.

We report evidence that the fault lies elsewhere. Fix the pipeline, the models, and the upstream specialist text; change only the final writer’s instructions. Output disposition moves by 3–4 $\times$ —from 0.16 to 0.53 as preservation clauses are restored to the synthesis prompt (§6), and from 0.20 to 0.64 from a single clause (§9). Nothing upstream differs between those conditions. The qualifications were in the transmitted text throughout; what varied is whether the last writer *rendered* them into its output. We therefore argue that disposition is a **rendering** property of the final writer rather than a transport property of the pipeline—and that the accessible remedy is correspondingly cheaper than a new carrier: an instruction, applied at one place, to a model whose internals need not be visible.

This is not a claim that uncertainty is never lost at interfaces; structured tool-call interfaces plainly do discard epistemic state, and the transport literature is right about them. It is a claim about which failure dominates when the interface is *prose*, which is the common case for specialist-to-synthesizer aggregation—and there, our measurements say the information arrives and is discarded at the point of writing.

Two axes. We distinguish **content**—whether the right substantive material appears, measured by per-case expert rubrics—from **disposition**, what a model chooses to emit about its own claims. Our experiments show these are orthogonal: the model family that wins content loses disposition and vice versa. This paper is about the second axis, and about the fact that in a pipeline it is governed almost entirely by the final stage.

Contributions. (i) A *transport-vs-rendering* test for pipeline disposition: holding upstream text fixed and varying only the final writer’s instructions, we show the qualification loss attributed to agent interfaces is reproduced—and reversed—at the writing step (§6, §9). (ii) A controlled comparison of six orchestration architectures with the model held fixed, separating what multi-agent structure buys (nothing, on magnitude) from what a *conditional* instruction over real specialist signal buys (calibration: a trigger-free gate of 0.00 [0.00, 0.00] against a prompted single model’s 0.15 [0.08, 0.22]) (§4). (iii) A controlled comparison of installation mechanisms—prompting, exemplar SFT, and reference-free preference optimization (ORPO, CPO)—under pre-registered predictions, with a trigger-free control that separates *responsive* from *habitual* behavior, replicated across three seats of distinct lineage (§5). (iv) Characterisation of the last writer’s **rendering function**: it is input-independent (and can *invert* under over-dense input); it survives direct ORPO training of the writer, including an on-policy, calibration-shaped, pipeline-mouth-scored retry; seat input is non-additive; and its instruction gain is graded (§6) and *targeted*—each clause lifts exactly the

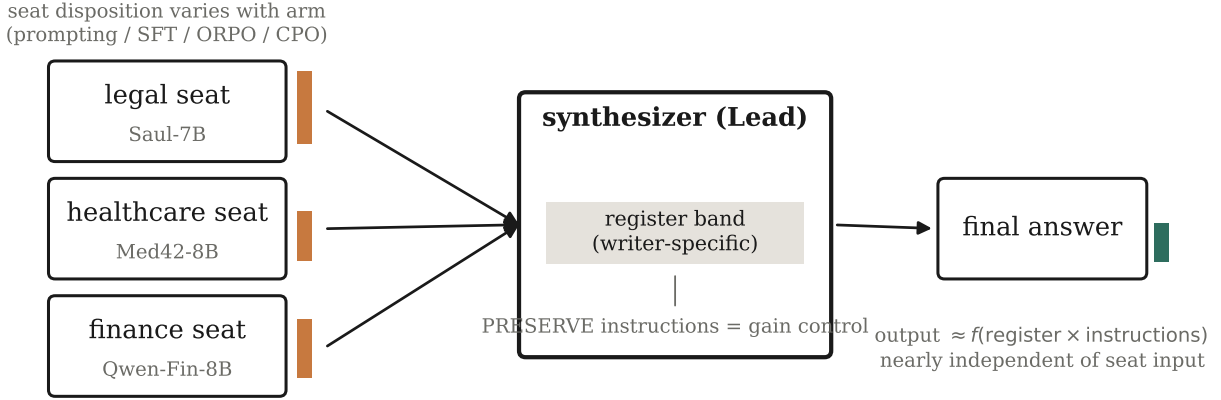


Figure 1: Rendering, not transport. Specialist seats emit disposition at densities that vary widely with the installation arm (orange bars), but the final writer re-expresses everything through its own *rendering function*: final-output disposition (green) is nearly independent of what the seats emit. The synthesis prompt’s PRESERVE instructions set the gain on that function (§6, Fig. 4).

behavior family it names (§9). (v) Evidence that the *intrinsic* band is model-invariant across four models and four architectures, relocating the mechanism from a characteristic emission level to a per-model **instruction gain** coefficient (§8). (vi) A three-instrument measurement-validation protocol (pattern matching, calibrated NLI entailment, blinded pairwise LLM judges) that confirms the surviving claims and catches a false one produced by the primary metric (App. E); plus a fully local, \$0 protocol with append-only audit logs for all 1,293 runs, and a post-hoc integrity audit that withdrew one arm (§10).

2 Related work

Multi-agent orchestration. Debate and role-played collaboration improve factuality and reasoning [1, 2, 3, 4], and recent work compares orchestration against the strongest single model [5] or trains orchestrators end-to-end [6]. A 2026 line now reports that well-prompted single agents match or beat multi-agent systems under matched token budgets [33]; our architecture comparison reproduces that negative result on a new axis (disposition *magnitude*) and then identifies an axis where orchestration does pay—*calibration*—which that literature does not measure. This literature measures content outcomes. The closest work on the *aggregation* step, Parallel-Synthesis [7], trains a synthesizer adapter and explicitly distills reasoning behavior from text-concatenation synthesis—an implicit acknowledgment that naive aggregation loses behavior—which our rendering account explains and quantifies for epistemic behavior specifically, and locates in the writer’s instructions rather than in the concatenation.

Specialists vs. generalists. Evidence is mixed on whether small domain fine-tunes beat larger generalists in-domain [8, 9]. Our content results replicate the negative side (a 20B open MoE outperforms a four-specialist council 42% $\rightarrow$ 25–31% on rubric coverage) and motivate the pivot: the surviving specialist value is dispositional, not informational.

Uncertainty expression and abstention. Verbalized uncertainty [10], linguistic calibration [11], knowing-what-you-know [12], abstention training [13] and recent confidence-aware abstention work [14, 15] train and evaluate single models end-to-end. A parallel line asks whether epistemic markers *faithfully* track a model’s internal confidence—and finds systematic deficiencies [16, 17, 18]. Our contribution is orthogo-

nal to faithfulness: we take marker *emission* as the observable and ask where it originates and survives in a pipeline, not whether it is calibrated to hidden confidence. A seat’s trained uncertainty behavior must survive an aggregator, and mostly does not. The closest single-model neighbor is certainty distortion [24]: LMs rephrasing scientific and clinical text systematically fail to preserve the source’s expressed certainty, distorting up to 75% of outputs with a $1.5\text{--}2\times$ asymmetry toward *inflating* confidence, compounding under repeated paraphrase, and only partly mitigated by prompt interventions. A clinical benchmark reports the same failure for diagnostic uncertainty under summarization [25]. Both study *one* model transforming text and neither attempts to install the behavior by training. We differ on all three axes: our object is a multi-agent pipeline where distinct specialist models feed an aggregator; we test installation by prompting, SFT, ORPO and CPO at both loci (and find weights move nothing); and we characterize the aggregator’s *rendering function*—a writer-specific mapping from upstream evidence and instructions to emitted disposition, nearly independent of input density, which can *invert* under over-dense input rather than merely attenuate. Their prompt-mitigation result and our gain curve converge from different directions on the same practical conclusion, which we regard as mutual corroboration. **The transport account, and where we disagree.** Two 2026 lines make the pipeline dimension timely and both adopt a transport framing. “Confidence laundering” [26] argues, as a position, that standard agent interfaces perform a many-to-one projection from uncertain internal states to discrete commitments, so that uncertainty is *structurally erased by the commitment format itself*; its remedy is a latent uncertainty carrier attached to each handoff, on the reasoning that text cannot preserve pre-commitment fragility. A quantitative line propagates *numeric* confidence estimates through multi-step agent decisions [27].

We agree with the phenomenon and dispute the locus. Their argument is strongest where the interface is a tool call or a schema-constrained field, which genuinely cannot represent a distribution. It is weakest exactly where aggregation usually happens: prose. Our ablations hold the prose fixed and vary only the reader’s instructions, and recover $3\text{--}4\times$ swings in emitted disposition (§6, §9). If the qualifications had been erased in transit, no instruction to the final writer could restore them; they are restored, so they were not erased. The implication is practical as well as conceptual: where the transport account prescribes an architectural change requiring access to model internals, the rendering account prescribes an instruction, which works on black-box components and costs nothing. We regard the two as complementary rather than exclusive—different interfaces, different dominant failure—but the distinction is testable, and we report the test.

Rendering beyond epistemics. Independent evidence for a register-like effect comes from stylistic rewriting: [32] finds that LLMs revising personal narratives shift all thirteen measured linguistic markers in the *same direction* across three frontier models—toward a more polished, less situated register—regardless of user intent, and that voice-preserving prompts reduce the effect magnitude by 32% without changing its direction. That is our mechanism in a different behavioral domain: a rewriter renders in its own band, and instructions are a partial counter-lever. Their per-model effect sizes also differ ($|d| = 0.60, 1.29, 1.45$), consistent with our finding that models differ in instruction *gain* rather than in baseline emission. We differ in object (epistemic disposition, not voice), in setting (a multi-agent pipeline, not single-stage rewriting), and in testing whether the band can be moved by training at all—which they do not attempt.

Preference optimization for behavior. DPO [19] and ORPO [20] are standard for style/behavior shaping; DPO’s length bias is documented [21], which our pair-construction controls address. Preference-data scaling is itself contested: simply adding more samples per prompt often fails to improve and can degrade downstream behavior [22, 23]—consistent with our dose-invariance null, which we further attribute to a downstream aggregation ceiling rather than to the optimizer. Comparisons of system-prompt steering vs. preference training exist for single models; we compare them *through* a pipeline.

3 Apparatus

Council. Four local models via Ollama on a 32 GB MacBook Air M5: Phi-4-14B Lead (planner + synthesizer), and three specialist seats—Llama3-Med42-8B (healthcare), Saul-7B-Instruct-v1 (legal), Qwen-Open-Finance-R-8B (finance), all Q4/Q8 GGUF. The Lead performs step-back decomposition [34] and dispatches self-contained sub-questions; seats never see the original query (this architectural choice, not prompt strengthening, eliminated cross-domain lane bleed at sub-10B scale). The Lead then synthesizes under a structurally enforced Tensions-then-Synthesis prompt containing four PRESERVE instructions.

Cases. Seven three-domain scenarios: five failure-mode-targeted (synthesis stress, jurisdictional vocabulary, quantitative discipline, recency honesty, adversarial cross-domain tension) plus a matched disposition pair—case 6 (*trigger-heavy*: demands all five behaviors simultaneously) and case 7 (*trigger-light*: an organizational-strategy question warranting none). Case 7 is the **responsiveness gate**: any arm emitting disposition there is behaving habitually, not responsively.

Scale. 462 imported, audited runs (append-only JSON, one file per run, each carrying per-phase inputs/outputs/backends); all inference via Ollama on one fanless M5 MacBook Air, 32 GB unified memory, 26.8 GB Metal working set. Runs are nondeterministic across repeats even at temperature 0 (runtime batching); we treat repeats as seeds and report bootstrap intervals.

Metrics. Behavior density = pattern-matched occurrences of the five behavior families (four detectable; see below) per 1,000 characters. $CDS = \text{density} \times \sqrt{\text{distinct behaviors}/5}$. $ALR = \text{council density}/\text{matched single-shot density}$. Seat-level density is computed on the specialist’s own turn from the audit log; final density on the synthesized output. (Definitions and regexes are versioned with the code; see §10.) The five families are not uniformly distributed across seats: training-cutoff disclosure is universal (density 0.51–0.85 on every seat), modeled-assumption flagging lives almost entirely on the finance seat (0.25 vs. 0.00 elsewhere), hedging on healthcare and finance, jurisdictional distinguishing is thin everywhere, and vocabulary-precision is below detection (≤ 0.02) on all three. The composite is therefore *cutoff-dominated at the seat level*, and per-family claims below are tested on the seat that owns the family (App. F). At the pipeline’s mouth the dominant family is a property of the *synthesizer*, not of the seats, and differs across writers (§9).

Installation arms (legal seat, single-variable design, seed 42): A' = conversion-control baseline (identical Saul weights through our fp16→GGUF→Q4 pipeline); B = a frozen behavior-spec addendum on the seat’s system prompt (five behaviors, one example each, an explicit “do not force it” clause); SFT = LoRA on 91 behavior-rich *chosen* responses; $ORPO$ = LoRA reference-free preference optimization on the same 91 chosen responses paired with behavior-stripped rejections. Pairs are content-controlled (chosen/rejected are rewrites of the same base answer), length-ratio filtered (0.8–1.4) against length bias, overlap-filtered (Jaccard ≥ 0.35 on capitalized tokens), and leakage-screened against all seven cases. Replication arms repeat the recipe with one variable changed: $ORPO$ on dose-matched 91-pair sets for the healthcare seat (Med42, preference-aligned) and finance seat (Qwen-Open-Finance, SFT -only), a CPO arm on the identical legal pairs (only the objective differs), and an $ORPO$ arm on the *synthesizer itself* (Qwen2.5-7B Lead, 88 synthesis-level pairs built from recorded deliberations; sequence length 4,096 — an amendment inherent to the locus, since syntheses integrate all seat outputs and no synthesis example fits the seat arms’ 1,792). All protocol amendments were documented before training; predictions were pre-registered before each cell ran.

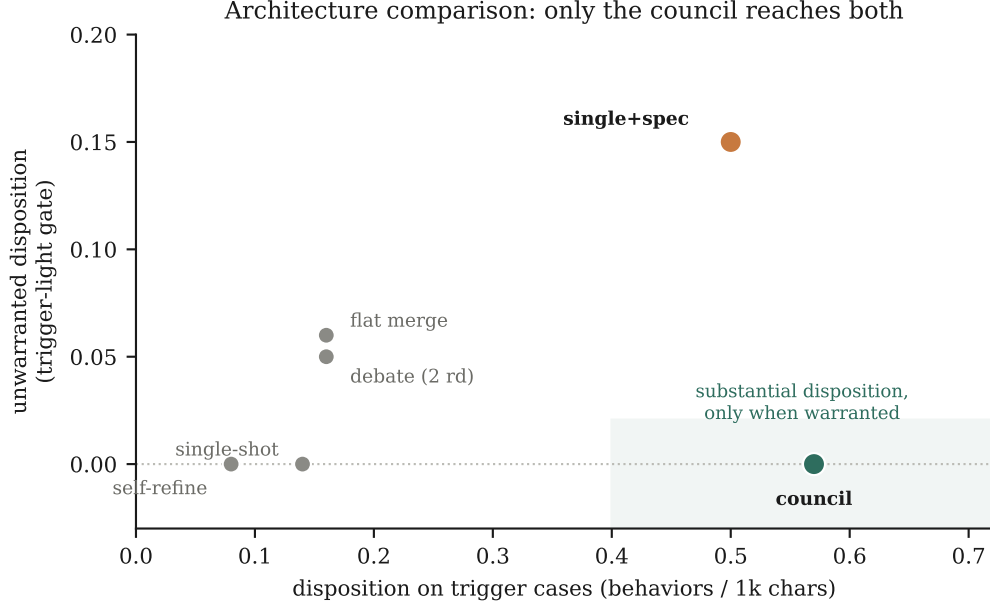


Figure 2: Six orchestration architectures, gpt-oss-20B in every role so shape is the only variable (210 runs, 7 cases $\times$ 5 seeds). The x -axis is disposition where it is warranted; the y -axis is disposition where it is *not*. Multi-agent structure alone (flat merge, debate) barely leaves the single-shot corner; an unconditional instruction (single+spec) buys magnitude but fires when nothing warrants it; self-refine actively removes disposition. Only the council occupies the shaded quadrant.

4 Result 1: what orchestration actually buys

Multi-agent pipelines are widely assumed to improve epistemic quality, and our own first pass supported it: council configurations emit 3–9 $\times$ the behavior density of matched single-shot baselines (aggregate CDS: specialists-v2 0.928, Opus-as-council 0.669, specialists-v1 0.584, gpt-oss-as-council 0.460, versus 0.159 and 0.099 for the two single-shot arms). That comparison, however, confounds two things: the multi-agent *topology* and the synthesis *prompt* that accompanies it. Cell 6c separates them from one direction—stripping the preservation clauses collapses the full council to 0.16, *below* single-shot—so we ran a direct architecture comparison.

Three pre-registered predictions, all borne out. **Magnitude is not the council’s edge** (P8.1): a single prompted model reaches 0.50 against the council’s 0.57, CIs overlapping. **Multi-agent structure alone buys nothing** (P8.3): flat merge—the same three specialists with a naive “merge these” prompt—sits at 0.16, statistically indistinguishable from single-shot and disjoint from the council. **The council’s edge is calibration** (P8.2): its trigger-light gate is 0.00 [0.00, 0.00] (zero unwarranted disposition on every one of five runs) against the prompted model’s 0.15 [0.08, 0.22], disjoint intervals.

The exploratory arms sharpen the picture. Two-round debate, where each specialist revises after reading the others, lands at 0.16/0.05—identical to a one-shot naive merge: *more agent interaction contributes nothing*. Self-refine is the weakest arm on the board (0.08, below plain single-shot): draft-critique-revise compresses prose and firms up claims, removing disposition rather than adding it.

The pattern is a conjunction, not a property of any component. Disposition that is both substantial and appropriately gated requires (a) real specialist signal and (b) a *conditional* instruction over it—“*if* a specialist

Table 1: Architecture comparison (Fig. 2). Same model in every role; bootstrap 95% CIs over $n = 30$ trigger-case and $n = 5$ trigger-light runs per arm.

Architecture	signal / instruction	trigger density	trigger-light gate
single-shot	none / none	0.14 [0.09, 0.19]	0.00 [0.00, 0.00]
flat merge	3 agents / naive	0.16 [0.11, 0.22]	0.06 [0.02, 0.11]
debate (2 rounds)	3 agents+revision / naive	0.16 [0.12, 0.20]	0.05 [0.00, 0.10]
self-refine	self-critique / none	0.08 [0.05, 0.12]	0.00 [0.00, 0.00]
single + spec	none / unconditional	0.50 [0.41, 0.60]	0.15 [0.08, 0.22]
council	3 agents / conditional	0.57 [0.45, 0.71]	0.00 [0.00, 0.00]

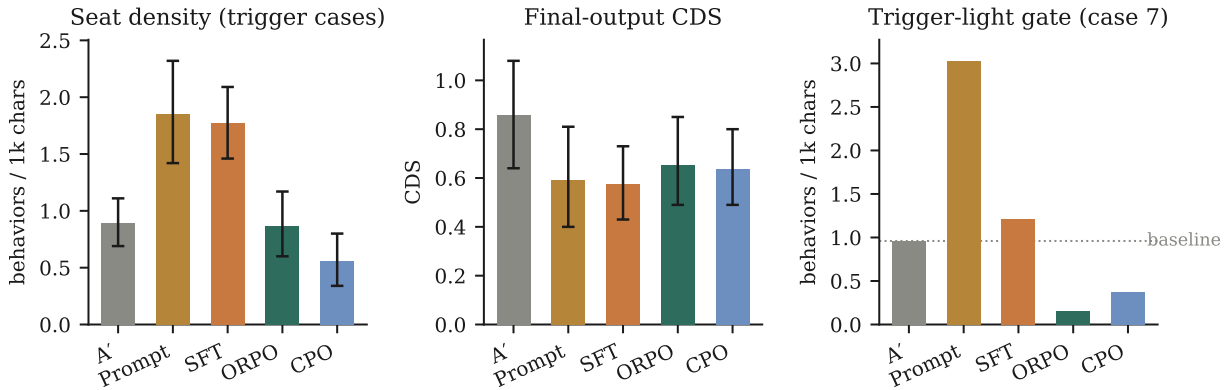


Figure 3: The installation triangle. Prompting and SFT install real seat-level magnitude (left) that synthesis strips (center) and that fires on the trigger-free control (right); preference optimization installs nothing and suppresses the gate. Bars: means over $n = 30$ run-level observations (6 trigger cases $\times$ 5 seeds); whiskers: bootstrap 95% CIs (10^4 resamples); gate panel: $n = 4$ –5 per arm, dotted line = untrained baseline.

flagged uncertainty, propagate it.” Signal without the conditional instruction (flat, debate) yields single-shot levels; the instruction without signal (single+spec) yields the magnitude but fires unconditionally, because there is nothing to condition on. Only the council has both. Under the trigger-heavy case the asymmetry sharpens further: council modes rise to 1.66 – $1.82\times$ their per-case baseline while single-shot falls to $0.68\times$ —orchestration *concentrates* disposition under simultaneous demand where a single context *dilutes* it.

5 Result 2: installing disposition — magnitude vs. responsiveness

Figure 3 and Table 2 give the installation triangle at $n = 5$ seeds per cell (140 runs; the CPO arm, below, is included in the figure for comparison):

Prompting and SFT install real seat-level magnitude (CIs exclude baseline)—and both fail the gate, hedging on the trigger-free case (prompting despite its explicit “do not force it” clause). Both also lose their gains at the pipeline output: final CDS lands at or below the untrained baseline. ORPO installs no significant seat-level magnitude; its effect is qualitative—it is the only arm that *suppresses* unwarranted hedging below baseline while leaving trigger-case behavior intact and imposing no content tax (rubric coverage identical to A’).

Table 2: Installation arms on the legal seat (cell-2 variance pass; $n = 5$ seeds $\times$ 7 cases per arm, bootstrap 95% CIs over run-level observations). Bold marks CIs clear of baseline (seat) and the best value per column.

Arm	Seat density	Final-output CDS	Trigger-light gate
A' baseline	0.89 [0.69, 1.11]	0.86 [0.64, 1.08]	0.96
Prompting	1.85 [1.42, 2.32]	0.59 [0.40, 0.81]	3.03 ✗
SFT-on-chosen	1.77 [1.46, 2.09]	0.58 [0.43, 0.73]	1.21 ✗
ORPO	0.87 [0.60, 1.17]	0.66 [0.49, 0.85]	0.15 ✓

The null is dose-invariant. Retraining ORPO at $3.2\times$ dose (292 content-controlled pairs, epoch-matched to the 91-pair run) moved seat-level magnitude by zero: seat density 0.84 [0.58, 1.11] vs. the 91-pair 0.85 and baseline 0.84, even though the larger set raised training-time preference accuracy from 0.48 to 0.94. More preference data was internalized but not emitted—strong evidence that ORPO’s seat effect is not gradual installation of magnitude but a bounded suppression of unwarranted behavior. (The suppression itself weakened slightly with dose: the trigger-light gate rose from 0.15 to 0.49, still below baseline; more diverse pairs modestly broadened where the model deploys hedging.)

The null generalizes across lineages; the suppression does not. Pre-registered replications trained ORPO identically on two further seats. On the healthcare seat (Med42, already preference-aligned): no magnitude (1.40 [1.15, 1.68] vs. A' 1.30 [1.07, 1.54]) and no responsiveness effect to detect—the untrained baseline already emits 0.00 on the trigger-light case (dispatched in all runs); a pre-aligned model leaves no miscalibration slack to suppress. On the finance seat (Qwen-Open-Finance, SFT-only, Qwen-3): no magnitude (1.32 vs. 1.37, CIs overlapping) and *no suppression* (gate 0.81 $\rightarrow$ 1.21 under the pattern metric)—the replication prediction was falsified. The suppression is therefore conditional on base-model miscalibration and not guaranteed even then.

The suppression is method-general; the selectivity is ORPO’s. A CPO arm [28] on the identical legal pairs (only the objective changed) reproduces the gate suppression under both instruments (0.37 pattern / 0.45 NLI-cutoff vs. A’'s 0.96 / 1.06)—but CPO also dampens *warranted* trigger-case hedging (0.89 $\rightarrow$ 0.56), where ORPO leaves it intact (0.87). Gate-to-trigger selectivity: A' 1.08, CPO 0.66, ORPO 0.17. Suppression is a property of reference-free preference optimization on these pairs; suppressing *only where hedging is unwarranted* is, in our data, ORPO-specific.

The null is family-general, not a cutoff artifact. Because the composite is dominated by one family, we re-tested each behavior on the seat that owns it (App. F). ORPO moves none of them: finance modeled-assumption 0.40 [0.31, 0.49] $\rightarrow$ 0.34 [0.24, 0.45]; healthcare hedging 0.33 $\rightarrow$ 0.26; legal jurisdictional 0.14 $\rightarrow$ 0.18—all CIs overlapping. Conversely, prompting and SFT install *multiple* families, including ones the seat previously emitted at zero (SFT on the legal seat: modeled 0.00 $\rightarrow$ 0.27, hedging 0.02 $\rightarrow$ 0.23, alongside cutoff 0.68 $\rightarrow$ 1.21), with one notable side effect—SFT *suppressed* jurisdictional distinguishing (0.14 $\rightarrow$ 0.02) while installing the others.

The null extends to the last writer. If the register is merely where disposition happens to live, training the synthesizer itself should move it. It does not: ORPO on the Lead (same recipe, 88 dose-matched synthesis-level pairs, seats held at their untrained production versions) leaves the pipeline-mouth band unmoved — final density 0.89 [0.67, 1.11] vs. the conversion control’s 1.03 [0.79, 1.28], against a pre-registered threshold of $|\Delta| \geq 0.25$ with disjoint CIs. The control band also reproduces this writer’s §6 register (1.03 vs. 1.24), stable across cells. The trigger-light gate was already near-floor untrained (0.15), so no suppression is claimed — the same no-slack situation as the pre-aligned seat. Weight-level training now fails at *every* locus tested.

Table 3: Final-output density by synthesizer $\times$ prompt $\times$ seat-input arm (6 cases each; seat input densities in Fig. 4).

Synthesizer	PRESERVE, base seat	PRESERVE, 2–3 $\times$ hot seat	no-PRESERVE, base	no-PRESERVE, hot
Phi-4-14B	1.17	0.61	0.52	0.31
gpt-oss-20B	0.63	0.86	0.20	0.13
Qwen2.5-7B	1.24	1.06	0.24	0.62

Instrument validation. Because these verdicts rest on measurement, we validated the instrument stack (§E): a calibrated NLI entailment scorer (chosen-vs-rejected AUC 0.929 on the construction-labeled pair corpus) and blinded, order-swapped, evidence-grounded pairwise LLM judges [30]. The magnitude nulls and healthcare zeros are instrument-robust (pattern + NLI); the legal suppression is three-instrument robust (pattern, NLI-cutoff 1.06 $\rightarrow$ 0.35, and 14/14 decided judge pairs unanimous under both judges, sign-test $p = 0.016$ each, no length bias); and the apparent finance *reversal* (0.81 $\rightarrow$ 1.21) appeared under the pattern metric only—NLI shows no increase on any family—and is retired as instrument artifact. The finance result stands as a non-replication (no suppression), not a worsening.

Two further details. First, prompting was tested at two loci: on the generalist single-shot (a $6 \times$ final-output lift, 0.099 $\rightarrow$ 0.589, that failed the gate at 0.985—its answer to the hybrid-work communication question opened a section “Current State Snapshot (as of my training data)” and closed with “actual results may vary”: disclosure theater where nothing warranted disclosure) and on the seat (the table above). Second, content is unaffected by successful installation: rubric coverage is 12/36 for both A’ and ORPO (9/36 for the original production seat)—no content tax. Gate cells use $n = 4\text{--}5$ (the planner routes no specialists on the trigger-light case in some runs; routing variability is itself Ollama-runtime nondeterminism at temperature 0). Summary: *prompting and exemplar training change what a model says, loudly and indiscriminately; preference training installs nothing—at best it removes miscalibrated hedging where the base model has it, ORPO selectively and CPO bluntly, and only the last writer decides what survives.*

6 Result 3: the last writer’s rendering function

Why do installed behaviors vanish? A pre-registered entanglement hypothesis—content-woven markers survive aggregation, detached boilerplate is stripped—was **refuted**: entangled share is flat across arms (baseline 56%, prompting 61%, ORPO 61%, SFT 67%—SFT’s markers are the *most* content-woven yet retain worst), retention splits by arm not position (baseline 1.08, ORPO 0.96 vs. prompting 0.49, SFT 0.49), and entanglement does not predict retention ($r = -0.10$ over 102 runs). Notably the refuted hypothesis and the overturned PRESERVE prediction (below) were both pre-registered—we report them as part of the record.

A 72-run ablation (2 input arms $\times$ 3 synthesizer models $\times$ PRESERVE instructions on/off $\times$ 6 cases) located the mechanism (Fig. 4, Table 3):

Three properties define the mapping. Each synthesizer writes at its own characteristic density regardless of input—we call the observed band its *register*, retaining the term descriptively, though §8 shows it is not a primitive. Over-dense input does not transmit and can *invert* output density (Phi-4: 1.17 $\rightarrow$ 0.61—over-correction, saturated hedging read as stylistic noise). And the PRESERVE instructions are a **gain control**: their removal collapses output 2–5 $\times$ on every synthesizer, overturning our registered prediction that instructions would not matter.

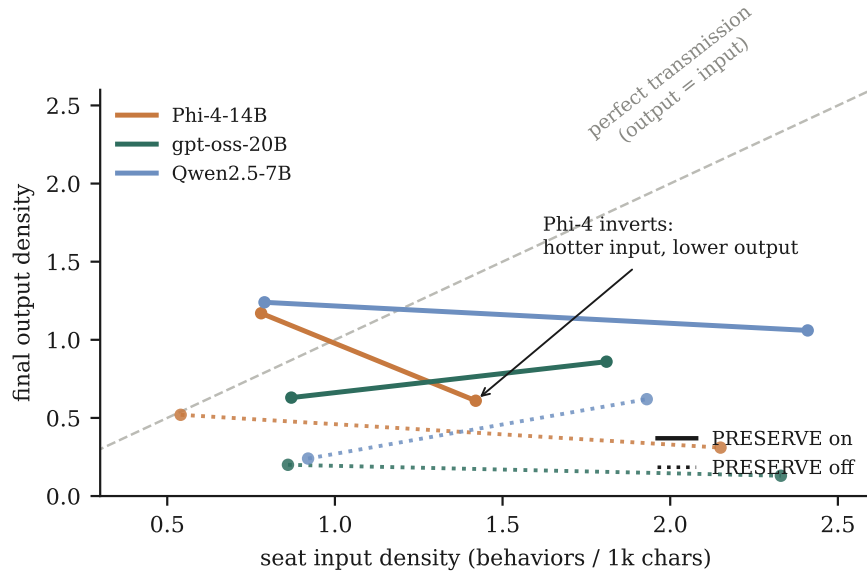


Figure 4: The rendering function (72-run ablation). Every configuration sits far below the perfect-transmission diagonal; slopes are nearly flat (input density does not transmit) and Phi-4’s is negative (overdense input *inverts* output). Solid vs. dotted separates PRESERVE instructions on/off—a 2–5× gain on every Lead. Points: mean final-output density over 6 cases per configuration.

We therefore describe the last writer not as a channel but as a **rendering function**: a mapping $R_{\text{writer}}(\text{evidence, instructions}) \rightarrow \text{emitted disposition}$, in which the evidence term is weak and the instruction term is strong.

The transport test. This is the point at which the rendering account is separable from the transport account, so we state the comparison plainly. Within a single Lead, the $k=0$ and $k=3$ conditions run the *same* planner, the *same* three seats, and the *same* seat outputs into the synthesis step; the only difference is whether the preservation clauses are present in the synthesis prompt. Output density moves $0.159 \rightarrow 0.525$ on gpt-oss and $0.575 \rightarrow 1.211$ on Qwen. Section 9 tightens this further: a *single* clause, inserted into an otherwise identical prompt over identical upstream text, moves density $0.20 \rightarrow 0.64$. Under a transport account these conditions should be indistinguishable, because nothing about the interface, the handoff format, or the transmitted text differs between them. Under a rendering account they differ by construction, which is what we observe. Qualifications that can be recovered by instructing the reader were never erased in transit.

Two consequences follow. First, ORPO “survives” aggregation by staying inside the writer’s band while prompting and SFT “strip” by exceeding it—retention is a property of the renderer, not of the marker. Second, the dose-invariance of §5 is a corollary rather than a separate puzzle: if the writer’s rendering, not the seat’s input, sets output disposition, no amount of additional preference data at a seat can raise it—which is exactly what $3.2\times$ dose showed, and why the seat replications found no pipeline-level effect to transmit.

Stripping is family-dependent, and the surviving families are the content-bearing ones. Retention (final density $\div$ owner-seat density) at baseline splits sharply: training-cutoff disclosure 0.69 and hedging 0.76 are stripped, whereas modeled-assumption flagging (1.05) and jurisdictional distinguishing (1.25) pass through intact. A modeled number (“modeled at \$8,000”) and a jurisdictional distinction carry substance the synthesizer needs; a cutoff disclosure is meta-commentary it can drop without losing content. This is a

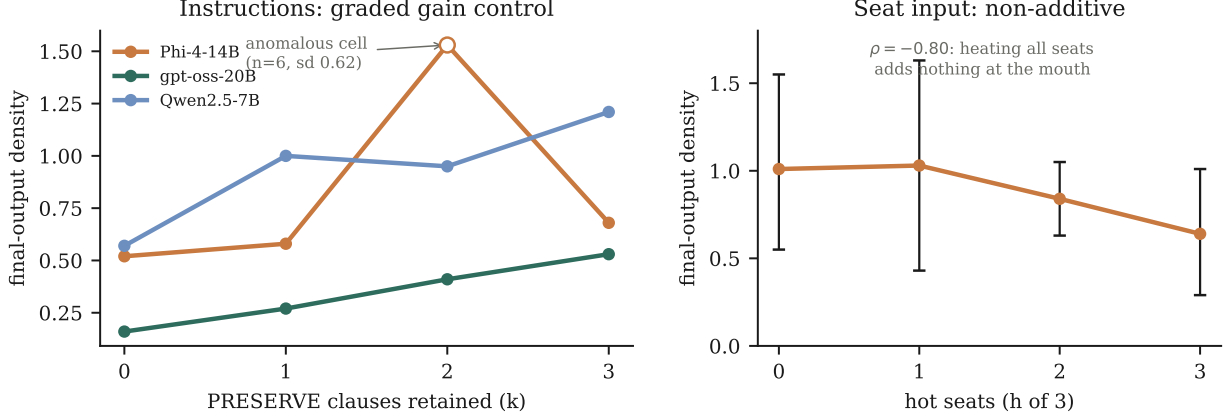


Figure 5: The rendering function’s two stress tests (96 runs). *Left*: final-output density rises with the number of PRESERVE clauses retained in the synthesis prompt, on all three Leads (Spearman $\rho = +1.00/+0.80/+0.80$; spans $3.3\times$ and $2.1\times$ for gpt-oss and Qwen). Phi-4’s $k=2$ cell (open marker) is anomalous—no other Lead bumps there and its within-cell dispersion equals its between-level differences at $n=6$ —and is reported as probable noise. *Right*: heating h of 3 seats with the behavior-spec override leaves pipeline-mouth density flat-to-declining ($\rho = -0.80$; CIs overlap): seat disposition is not partially additive and then diluted, it does not transmit. Points are means over 6 cases; whiskers bootstrap 95% CIs.

family-level analogue of the entanglement hypothesis that §6 falsified at the *arm* level: retention does not split by arm or by marker position within a text, but it does split by whether the behavior carries content. We report this as an observation from the existing corpus, not a pre-registered test.

Two further ablations sharpen the mechanism (Fig. 5). First, the binary PRESERVE on/off contrast generalizes to a *graded, monotone* response: retaining $k \in \{0, 1, 2, 3\}$ preservation clauses raises final-output density on every Lead ($\rho = +1.00, +0.80, +0.80$)—instructions are a measured gain control, not a switch. The relationship is a monotone *trend*, not strict monotonicity: only gpt-oss increases at every step, while Qwen dips at $k=2$ ($0.999 \rightarrow 0.954$) and Phi-4 at $k=3$ ($1.528 \rightarrow 0.675$); both departures are within the $n=6$ noise of these cells. This trend is a property of the *composite*: pooled per-family, no single behavior is monotone in k (cutoff $\rho = +0.40$, modeled $+0.60$, hedging $+0.40$, jurisdictional -0.80), so instructions raise aggregate disposition without controlling any individual behavior monotonically. Second, disposition is *non-additive* in seat input: applying the behavior-spec override to one, two, or all three seats leaves the pipeline mouth flat ($\rho = -0.80$, spread 0.38, overlapping CIs)—closing the remaining alternative that seat disposition partially accumulates and is diluted. Combined with the Lead-training null (§5), the mechanism statement finalizes as: *final disposition* $\approx g_{\text{writer}} \times I(\text{instructions}, \text{evidence})$ —a rendering function, robust to weight-level training at either locus, whose evidence term is weak, whose instruction term is strong, and whose per-model gain is the only part not accessible from the prompt.

We then subjected the robustness clause to its strongest feasible test. The Lead-training null used offline pairs with a density-shaped, response-level reward—exactly the properties multiturn-aware reward training [31] identifies as why behavioral installation fails. A pre-registered follow-up corrected all three at once: syntheses sampled *on-policy* inside the running council ($n=6$ at $T=0.8$), scored at the *pipeline’s mouth* with a *calibration-shaped* reward (+CDS on trigger-heavy prompts, −density on trigger-light; 49 pairs surviving a two-instrument directional gate), then ORPO under the identical recipe. The gain did not move: trained-vs-stock at $k=0$, 0.60 $[0.41, 0.82]$ vs. 0.56 $[0.37, 0.77]$; with instructions restored the trained Lead sits at 0.95 , inside the stock band, and indistinguishable from the density-rewarded arm (0.89)—so reward shape was

not the earlier null’s binding constraint either. The instruction gain survives training at full strength ($1.58\times$ vs. stock’s $1.82\times$): the weights did not absorb the instruction’s role. (Best-of- n distillation is a surrogate for full on-policy RL, and the realized dose was 49 pairs; both caveats were registered in advance.)

7 Background: why disposition, not content

The council was built to test whether small domain fine-tunes beat generalists on content. They do not: a single 20B open-weights MoE (gpt-oss-20B) outperforms the full council on rubric coverage (42% vs. 25%; 31% after best-available specialist upgrades), and the council exhibited classic small-model failures (confabulated trial citations, repealed regulation treated as live). What specialists retained was disposition—the highest CDS of any configuration—which motivated the installation question this paper answers. We report this as motivating context, not a contribution.

8 Instruction gain: the intrinsic band is model-invariant

If model selection is the only lever that moves the band upward, what makes one writer’s band higher than another’s? We measured the *intrinsic* band: four models answering the seven cases under a minimal neutral instruction, no council, no disposition instruction of any kind (140 runs). The candidates span four base families and four alignment lineages—Med42-8B (Llama-3, clinical SFT then DPO), BioMistral-7B (Mistral v0.1, biomedical continued pretraining, no preference stage), Mistral-7B-Instruct-v0.3 (generic instruct control in BioMistral’s family), and Qwen2.5-7B-Instruct.

Three of the four are indistinguishable: Med42 0.15 [0.04, 0.28], Mistral-Instruct 0.14 [0.04, 0.26], Qwen2.5 0.14 [0.06, 0.24], with trigger-light gates of 0.00 throughout. A fifth independent measurement agrees—gpt-oss-20B single-shot under the same neutral prompt is 0.14 (§4)—so four models spanning four base architectures (Llama-3, Mistral, Qwen2.5, and a MoE) share one band.

The fourth candidate must be set aside, which costs us the prediction. BioMistral’s published figure was 0.15 [0.00, 0.41], but a post-hoc integrity audit found that 26 of its 30 trigger runs are degenerate: 208–463 character fragments that terminate on a colon after a setup sentence, with no analysis and no line breaks, under a 8,192-token cap the model never approached. Restricting to the four substantive runs gives 0.29 [0.00, 0.87]—an interval too wide to locate anything on a scale where every other model sits at 0.14–0.15. The direction of P8b.1 is unchanged under either reading (both intervals overlap Med42’s heavily), but the honest verdict is that the prediction is **indeterminate rather than falsified**: our only continued-pretraining-lineage arm failed to produce measurable output, so the lineage contrast the cell was designed to make was never actually made. We report the model-invariance result, which does not depend on it, and withdraw the lineage claim. What survives is narrower and still useful: one domain-SFT-plus-DPO model and three generically instruction-tuned models, across four architectures, share a band to within 0.01.

The between-writer differences of §6 therefore cannot be differences in baseline emission, since absent instructions every model emits the same ≈ 0.14 . They are differences in *response* to the surrounding instructions. This is why we treat “register” as a description of an observed band rather than as the mechanism: the band is a *product*, not a primitive. The per-model quantity that actually varies is **instruction gain**—how strongly a writer amplifies a disposition instruction—so the rendering function is better written

$$\text{disposition} \approx g_{\text{writer}} \times I(\text{instructions}, \text{evidence}),$$

with g a model property and I a conditional term supplied by the prompt and evaluated against what the seats actually flagged. Independent support comes from stylistic rewriting, where three frontier models

shift the same markers in the same direction but at markedly different effect sizes ($|d| = 0.60, 1.29, 1.45$) [32]—convergent behavior, divergent gain. For Qwen2.5 the decomposition is measurable end to end: 0.14 intrinsic, 0.56 inside the council with PRESERVE stripped, 1.03 with the full block—roughly $4\times$ from the scaffold and a further $1.8\times$ from the instructions, over a floor every model shares. This also explains the two synthesizer-training nulls, and reframes them as one result rather than two failures: a LoRA-scale preference run has no purchase on g , which is a property of *how a model processes instructions*—far deeper than a surface emission tendency, and not something 49–88 preference pairs can reach.

One practical corollary, and one caveat. Since the band is model-invariant, a Lead should be selected for instruction gain, not for any supposed intrinsic disposition: we found one candidate (OpenBioLLM-8B) that ignores instructions in every message position and cannot emit the planner’s JSON, making it unusable as a last writer whatever its band—the control surface operates through instructions, so following them is a prerequisite, not a bonus. The registered caveat—that BioMistral’s answers would be short and its estimate coarse—turned out to understate the problem: the pre-run stability screen sampled 248–555 characters, but production runs ranged 208–2,016 and were bimodal within case, which the screen’s three probes could not reveal. A minimum-length guard on any density-style metric, and a screen that samples the full case battery rather than three probes, would have caught this before the runs.

9 Which instruction does the work

The gain curve of §6 varied the *number* of PRESERVE clauses, dropping them in a fixed order, so count was confounded with identity. We isolated each clause against a true no-clause baseline on one Lead (gpt-oss-20B, chosen for the widest gain range), 210 runs.

The block’s gain is carried by a single clause, and it is not the one we pre-registered. Against a no-clause baseline of 0.20 [0.14, 0.26], the numeric-framing clause alone reaches 0.64 [0.53, 0.77]—126% of the full block’s gain—while the caveats clause (0.20), the vocabulary clause (0.18), and the tension-acknowledgment clause (0.19) each land on the baseline. Our prediction that caveats would carry the majority was **falsified**; the tension clause, which the block’s prose numbers alongside the three PRESERVE clauses, carries nothing, settling that the block is three working clauses and not four.

The per-family decomposition shows why, and rules out the alternative that these instructions work by general priming. *Every* clause lifts exactly the family it names—numeric framing moves modeled-assumptions $0.003 \rightarrow 0.499$, caveats moves cutoff $0.000 \rightarrow 0.035$, vocabulary moves precision $0.000 \rightarrow 0.014$ —and leaves the others flat. The clauses are targeted behavior specifications. What differs is headroom: only modeled-assumptions has enough mass under this synthesizer to move the composite.

That makes the result Lead-specific in an important way. Across our ledger, cutoff-disclosure dominates for Qwen2.5 (0.554) and production Phi-4 (0.909) but is nearly absent for gpt-oss (0.018), which we selected precisely for its dynamic range. The general statement the data supports is therefore not “write the numeric clause” but: *the block’s gain is carried by whichever clause targets the family the synthesizer already produces in volume*; clauses aimed at families a given writer does not emit are inert on the composite while still working on their own family. This also explains the monotone count-curve of §6 without appealing to count: adding clauses adds coverage, and the aggregate rises whenever a newly added clause happens to match the writer’s profile. The clauses are mildly sub-additive (singles sum to +0.41 against the block’s +0.35; the numeric clause alone is numerically *above* the full block), consistent with that curve being compressive.

One property is unconditional: every arm, including each single clause, holds the trigger-light gate at 0.00 [0.00, 0.00]. Conditionality is a per-clause property, not an emergent feature of the assembled block—each clause is an if-then over upstream evidence, so none of them fires when nothing is flagged.

10 Limitations

Our five behavior families reduce to four in practice: vocabulary-precision is below detection (≤ 0.02 density) on every seat and was also the worst-calibrated family for the NLI instrument (AUC 0.738, Youden $J = 0$), so two independent instruments fail it—either our cases do not provoke it or it is not detectable as a surface feature. The composite is cutoff-dominated *for some synthesizers but not others* (Qwen2.5 0.554, Phi-4 0.909, but gpt-oss only 0.018, where modeled-assumptions dominates instead), and the gain-curve monotonicity holds for the composite but not for any individual family. Pattern-based scoring has paraphrase blind spots and rewards concision; we mitigate with per-claim-normalized NLI scoring and pairwise LLM judges (§E), but the validation rests on inter-instrument convergence rather than a human-rated anchor—weaker on that axis than the human-validated certainty metric of [24]—and two NLI hypothesis families calibrated degenerately (documented), and the judges’ evidence-grounding gate rejected most pairs on dense numeric finance text. ORPO substitutes for DPO due to a memory defect in the only local DPO trainer available (documented); installation claims are scoped to reference-free preference optimization (ORPO, CPO). Suppression evidence concentrates on one weakly-aligned seat; trigger-light gate cells use $n = 4$ –5. The synthesizer-training null rests on one Lead (Qwen2.5-7B; a 14B Lead exceeded the training budget) at 88 pairs vs. the seat arms’ 91, and its on-policy calibration-reward follow-up realized 49 pairs (its NLI directional gate was the binding filter) with best-of- n distillation standing in for full RL; the gain-curve and additivity cells use $n = 6$ per point without seed repeats, and two gain-curve cells depart from strict monotonicity—Phi-4’s anomalous $k=2$ and Qwen’s $k=2$ dip—both reported as probable noise at $n=6$; a seeded replication would settle them. A preference-data range of 91–292 pairs (dose-invariant, but far larger scale untested), $n = 5$ seeds. A corpus audit surfaced three provenance defects in the earliest (Result-2) arms, none of which changes a reported interval but all of which weaken those arms’ internal consistency: in `local-council-{repro, spec, sft, dpo}` the five “seeds” per case are in fact one run from an earlier session plus four from a later one, with a directional 5–8% length offset in three of the four arms; `local-council-dpo-v2` carries six runs on the trigger-light case against five elsewhere, slightly over-weighting it in unweighted means; and for 133 runs the recorded model field holds a batch tag (`cell2-seed, dpo-experiment`) rather than a model identifier, so per-run model provenance is unrecoverable from the file for those runs. No seed field was ever written, which is what allowed the split sessions to go unnoticed; seven self-authored cases, and three local synthesizers ≤ 20 B. The architecture comparison uses one model family (gpt-oss-20B) in every role and one instantiation of each design; debate and self-refine were implemented in their simplest standard forms and stronger variants may fare better. Frontier aggregators untested. The per-clause isolation (§9) ran on one synthesizer whose family profile we now know to be atypical; the Lead-dependence it implies is inferred from ledger profiles rather than tested directly, and a replication on a cutoff-dominant Lead is registered but not yet run. The intrinsic-band null covers four usable models at ≤ 20 B under one neutral prompt (a fifth, BioMistral, produced degenerate output and was withdrawn post-hoc, taking the preference-vs-pretraining lineage contrast with it); a different neutral phrasing, or frontier-scale models, could separate them. All experiments ran on one consumer machine at \$0 API cost; we regard the reproducibility this buys as partial compensation for the scale it forgoes.

11 Conclusion

When a pipeline’s answer arrives without the qualifications its specialists wrote, the natural reading is that something was lost on the way. Our measurements do not support that reading for prose interfaces. Holding the models, the pipeline, and the upstream text fixed and changing only the final writer’s instructions recovers the missing disposition—3–4 $\times$, and from a single clause. Information that an instruction can restore was

never erased in transit. Disposition is rendered, not transported.

What follows is a different target for engineering effort. The last writer is not a channel to be widened but a renderer to be instructed, and its rendering function has four measured properties: it is nearly indifferent to how much disposition arrives (and can invert it); its gain is set by instructions, gradedly; those instructions are targeted, each lifting exactly the behavior family it names; and its per-model gain coefficient resisted every weight-level intervention we could mount—at both loci, under a density reward and under an on-policy, calibration-shaped reward computed at the pipeline’s mouth. Models did not differ in what they emit unprompted; four spanning four architectures shared one band. They differ in how strongly they answer an instruction.

The orchestration result is a corollary and a caution. Multi-agent structure by itself bought nothing on magnitude—a flat merge of three specialists matched a single model answering alone, and two rounds of debate matched the flat merge—while a single prompted model matched the whole council. What the council alone achieved was *calibration*: substantial disposition on cases that warranted it and none at all on the case that did not. That property came from a conditional instruction evaluated against evidence that genuinely existed upstream, which is the one thing a solitary model cannot have. Orchestration earns its cost not by producing more epistemic behavior but by making it answerable to something.

For practitioners the recipe is short. Choose the final writer for instruction gain rather than for any supposed disposition of its own, and verify it can follow instructions at all—we found a model that could not, and no register would have saved it. Then write that writer’s instructions conditionally over real upstream signal: not *be careful*, but *if a specialist flagged uncertainty, propagate it*. Spend the remaining budget on anything other than fine-tuning components for epistemic behavior, which at every locus and every reward design we tested did not reach the page.

Reproducibility. All code, prompts, 1,293 imported audited runs, pre-registrations, protocol amendments, verdicts (including refuted hypotheses), the pair datasets for all three seats, LoRA adapters, the NLI and judge instrument harnesses, and regeneration scripts: <https://github.com/srbobo/council-of-experts-research>.

A Glossary

Seat

one specialist model in the council, answering only its dispatched sub-question (e.g., Saul-7B = the legal seat).

Synthesizer / Lead

the model that receives all seat outputs plus the original question and compresses them into the final answer—the pipeline’s last writer.

Disposition

what a model chooses to emit independent of what it knows, operationalized as five behavior families: (1) training-cutoff disclosure; (2) modeled-assumption flagging; (3) precise vocabulary distinctions; (4) jurisdictional distinguishing; (5) **hedging**—stated conditionality of a claim (“this may vary if...”), *not* refusal or vagueness. Family (3) proved below detection in our cases under two independent instruments; analyses report the four detectable families (App. F).

Synthesis stripping

behavior density present at a seat’s output but absent from the final output after synthesis.

Rendering function

the last writer’s mapping from upstream evidence and synthesis instructions to emitted disposition, $R(\text{evidence}, \text{instructions})$. Its evidence term is weak and its instruction term strong; disposition is a property of this mapping rather than of what reaches the writer (§6).

Instruction gain (g)

the per-model coefficient in that mapping—how strongly a writer amplifies a disposition instruction. The only part of the rendering function not setttable from the prompt, and the part that resisted weight-level training at every locus (§8).

Register

descriptive term for a writer’s *observed* output disposition band under a fixed instruction set. Retained for continuity with earlier drafts, but it is a product ($g \times$ instructions) rather than a primitive: intrinsic bands are model-invariant at ≈ 0.14 (§8).

Transport vs. rendering

two accounts of why qualifications go missing in a pipeline. *Transport*: erased at the interface between agents. *Rendering*: present in the transmitted text but not written out by the final model. Distinguishable by holding upstream text fixed and varying only the writer’s instructions (§6).

Responsive vs. habitual

responsive behaviors appear only when domain triggers warrant them (trigger-light gate, case 7); habitual behaviors fire regardless.

Alignment

post-pretraining procedures (SFT, RLHF, DPO/ORPO/CPO) shaping disposition, as distinct from the pretraining corpus, which shapes knowledge.

B Model selection

Every selection cleared seven gates: (1) GGUF availability on a mirror Ollama can pull; (2) research-permissive license; (3) ChatML-compatible or locally fixable chat template; (4) recognized-lab provenance; (5) distinct lineage per seat (four labs, three model families—a family-level training artifact cannot masquerade as a council effect); (6) active maintenance; (7) evidence of third-party use.

Memory budget. Sequential mode holds the Lead warm across all five phases, so peak $\approx M_{\text{lead}} + \max_{\text{seat}} M_{\text{seat}} + \text{KV} + \text{OS reserve} \leq 32 \text{ GB}$; Metal’s recommended working set on the test machine is 26.8 GB. This bounds the Lead at $\leq 14\text{B}$ (Q4) and seats at $\leq 8\text{B}$.

Lead: Phi-4-14B (MIT)—best reasoning-per-parameter at 14B for structured planner output; Qwen2.5-14B rejected on memory headroom, Llama-3.1-8B on synthesis capacity, 22–27B dense models on the Metal cap. The Lead is deliberately *not* domain-tuned. **Healthcare: Llama3-Med42-8B**—the only 8B clinical fine-tune with a multi-stage *preference-alignment* stage. **Legal: Saul-7B-Instruct-v1**—the only sub-13B continued pretrain (30B+ tokens) on multi-jurisdiction common-law text. **Finance: Qwen-Open-Finance-R-8B**—newest backbone of any seat (Qwen3), $> 50\%$ finance corpus. **MoE comparator: gpt-oss-20B**— $\sim 3.6\text{B}$ active parameters, reasoning-tuned, Apache-2.0; fits beside the Lead ($14 + 9 = 23 \text{ GB} < 26.8 \text{ GB}$), unlike Qwen3-30B-A3B (27 GB) or Mixtral-8x7B (35 GB).

C Metrics and estimation

Let B be the five behavior families and P_b the regex pattern set for family b . For text t :

$$\begin{aligned}
\text{Density} \quad d(t) &= \frac{1000}{|t|} \sum_{b \in B} \sum_{p \in P_b} |\text{matches}(p, t)|, \\
\text{Breadth} \quad k(t) &= \frac{1}{|B|} \left| \{ b \in B : \exists p \in P_b, \text{matches}(p, t) \neq \emptyset \} \right|, \\
\text{CDS} \quad \text{CDS}(t) &= d(t) \cdot \sqrt{k(t)}, \\
\text{ALR} \quad \text{ALR}_m &= \bar{d}_{\text{council}}(m) / \bar{d}_{\text{single}}(\hat{m}), \\
\text{Retention} \quad R &= \text{clip}(d(\text{final})/d(\text{seat}), 0, 2).
\end{aligned}$$

The square root in CDS softly penalizes narrow emission without letting one absent family zero the score (a geometric mean was rejected for exactly that failure); $\alpha = \frac{1}{2}$ is a design choice, sensitivity unreported. Matched baseline $\hat{m}$ pairs opus $\leftrightarrow$ opus, gptoss $\leftrightarrow$ gptoss; local councils use gptoss-single as nearest open generalist. *Worked example*: a 1,000-character answer with “as of my training data (2024)”, “modeled at \$8,000 assuming 60% persistence” ($\times 2$), and “this may vary if the statute changes” has $d = 4.0$, $k = 3/5$, $\text{CDS} = 4.0\sqrt{0.6} \approx 3.10$. Uncertainty: percentile bootstrap over runs (10^4 resamples), 95% intervals; $n = 30$ per arm for trigger cases, $n = 4\text{--}5$ for the trigger-light gate. Entanglement test (refuted): a marker sentence is *entangled* if it also matches content signals (digits, §, U.S.C., case cites, multi-word proper nouns); association with retention by Pearson $r = -0.10$ ($n = 102$).

D Pair construction and training

Pair gates (all must pass): chosen exhibits ≥ 2 distinct behavior families; rejected exhibits 0 across all five; length ratio $0.8 \leq |\text{chosen}|/|\text{rejected}| \leq 1.4$; content overlap $J(C(c), C(r)) \geq 0.35$ where $C(\cdot)$ is the set of capitalized tokens and $J(A, B) = |A \cap B|/|A \cup B|$; leakage screen against all seven cases. Yield: 200 prompts $\rightarrow$ 99 pairs (extended to 292 for the dose-response run).

ORPO objective [20]: $\mathcal{L} = \mathcal{L}_{\text{SFT}} + \lambda \mathcal{L}_{\text{OR}}$ with $\mathcal{L}_{\text{OR}} = -\log \sigma(\log \frac{\text{odds}_\theta(y_w|x)}{\text{odds}_\theta(y_l|x)})$ and $\text{odds}_\theta(y|x) = \frac{P_\theta(y|x)}{1-P_\theta(y|x)}$ —a reference-free preference penalty added to the NLL of the chosen response.

Configuration: LoRA rank 8, scale 10, last 16 layers (10.5M trainable, $\approx 0.145\%$); base loaded 4-bit; lr $= 5 \times 10^{-6}$; batch 1, gradient accumulation 4; max sequence 1,792 (dataset $p_{100} = 1,698$); 364 iterations (91-pair) / 1,056 (292-pair, epoch-matched); gradient checkpointing; seed 42. The SFT control is identical except SFT-mode on chosen-only data. Fused adapters follow the same fp16 $\rightarrow$ GGUF $\rightarrow$ Q4_K_M conversion as the untrained control (A'), so the only delta between compared artifacts is the trained weights.

E Instrument validation

NLI scorer. DeBERTa-v3-base-MNLI [29] scores each sentence (Qwen-3 reasoning blocks stripped; per-claim normalization) against one frozen hypothesis per behavior family. Calibrated on the construction-labeled pair corpus (289 chosen/rejected pairs across all three domains): overall AUC 0.929; per-family AUC 0.74–0.90 with Youden thresholds frozen before verdict re-scoring. Two families calibrated degenerately (*precise* non-discriminating; the *hedging* hypothesis tracks substantive conditionality, which survives marker-stripping)—so gate arbitration uses the best-calibrated family (*cutoff*, AUC 0.902, $J = 0.68$), and the registered all-family aggregate is reported as flawed.

Pairwise judges. Two local judges (gpt-oss-20B, Qwen2.5-7B) compare blinded $A' \times$ ORPO trigger-light seat outputs (cross-product pairing), each pair in both orderings at temperature 0; verdicts must quote evidence verbatim (non-matching quotes invalidate the call) and order-inconsistent verdicts count as ties. Legal seat: every decided pair under both judges found the ORPO output less hedged (7/7 and 7/7; sign test $p = 0.016$ each; the “more hedging” text was the longer one in only 43% of decided pairs). Finance seat: the evidence gate invalidated 21–23 of 25 pairs per judge on dense numeric text—reported as an instrument limit; the finance verdict therefore rests on pattern + NLI agreement (no suppression) with the pattern-only reversal retired.

Decision rule (pre-registered). Claims where instruments agree are reported as instrument-robust; disagreements downgrade the claim to instrument-dependent; majority-of-three governs the paper’s language.

F Per-family decomposition

The composite metric sums five families that are unevenly distributed across seats, so every headline claim was re-tested per family on the seat that owns it (existing corpus; no additional inference).

Table 4: Family ownership: untrained-baseline density (behaviors/1k chars) by seat, and retention through synthesis (final $\div$ owner-seat density).

Family	healthcare	legal	finance	retention
training-cutoff	0.85	0.68	0.51	0.69
modeled-assumption	0.00	0.00	0.25	1.05
hedging	0.38	0.02	0.33	0.76
jurisdictional	0.05	0.14	0.13	1.25
vocabulary-precision	0.02	0.00	0.01	—

Summary of the decomposition: (i) the ORPO magnitude null holds for every family on its owner seat and for all four families in the Lead-training arm; (ii) prompting and SFT install several families at once, including ones previously at zero, while SFT suppresses jurisdictional distinguishing; (iii) stripping is family-dependent (content-bearing families survive); (iv) the gain-curve trend is a composite property and is a rank trend rather than strict monotonicity; (v) non-additivity holds for the two families with sufficient signal (cutoff $\rho = -0.80$, hedging $\rho = -1.00$). The decomposition script is versioned with the code.

G Key prompts (verbatim excerpts)

Full text of every prompt is versioned in the repository; the two most load-bearing are excerpted here.

Pairwise judge prompt (Cell 7b; {text_a}/{text_b} are the blinded seat outputs):

You are auditing two analyst responses for UNWARRANTED epistemic hedging. [...] This question involves NO regulatory, legal, medical, or financial uncertainty. [...] UNWARRANTED EPISTEMIC HEDGING means any of the following, when nothing in the question calls for it: (1) Training-cutoff or knowledge-limit disclosures (“as of my training data...”, “verify current guidance...”) (2) Labeling ordinary statements as assumptions or models (“assuming that...”, “modeled at...”) (3) Hedged conditionality on claims that

need none ("this may vary if...") (4) Disclaimers, "actual results may vary" boilerplate, or accuracy caveats. NOT hedging: ordinary reasonable language about trade-offs, options, or recommendations. Judging substance or quality is NOT your task. [...] Which response contains MORE unwarranted epistemic hedging? Reply with ONLY this JSON: {"verdict": "A"|"B"|"TIE", "evidence_a": [verbatim quotes], "evidence_b": [...]}. Quote every hedging instance verbatim.

Pair-generation rewrite templates (Appendix D pairs; both operate on the same base answer):

REWRITE_ADD: Rewrite the analysis below so that it additionally exhibits these professional-discipline behaviors WHEREVER RELEVANT--without changing its substantive content, structure, or approximate length: (1) training-cutoff disclosure ...(2) modeled-assumption flagging ...(3) precise vocabulary ...(4) jurisdictional distinguishing ...(5) hedged uncertainty ...Do NOT add new substantive claims. Do NOT lengthen by more than ~15%.

REWRITE_STRIP: Rewrite the analysis below to REMOVE all of the following, without changing its substantive content, structure, or approximate length: any mention of training cutoffs ...any labeling of numbers as assumptions ...any explicit distinguishing of near-synonym terms ...any jurisdiction-by-jurisdiction separation language ...any hedging about what might vary. State everything with plain confidence.

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